Problem-2

A sensor → A, B, C } 8 bit readings

order: A → B → C

Processing unit (Receiver)

When processing unit is busy :- It cannot accept already.

Handshake rule

① VALID must not depend on READY. The Tx offers data first

② While READY = 0, hold VALID and DATA_OUT unchanged.

③ When a transfer happens, the next reading appears in very next cycle.

RESET) sync & active high

RESET = 1, controller → idle state with Valid = 0 and DONE = 0.

START) To begin the cycle (one-cycle req)
only given while the controller is idle.

READY) High means the receiver can accept data at the next rising edge. (Low means it is busy)

DATA_OUT) The reading concurrently being sent.

VALID → High means DATA-OUT contains a reading that is ready to be transferred.

DONE → High for exactly one clock cycle after all three readings have been accepted.

Designing the FSM

The timing convention says that if a transfer occurs, the FSM moves to its next state on that edge, and the new output appears in the following cycle.

⇒ Output changes only if the state changes, which is Moore behaviour

Our FSM sends 8 bit readings A, B and C to a receiver. The receiver may be busy, so the transmitter must wait without losing or repeating any reading.

So we will choose our states based on the readings that we have to transmit to the receiver.

* States *

S_0 ⇒ IDLE
Here, Valid = 0, Done = 0

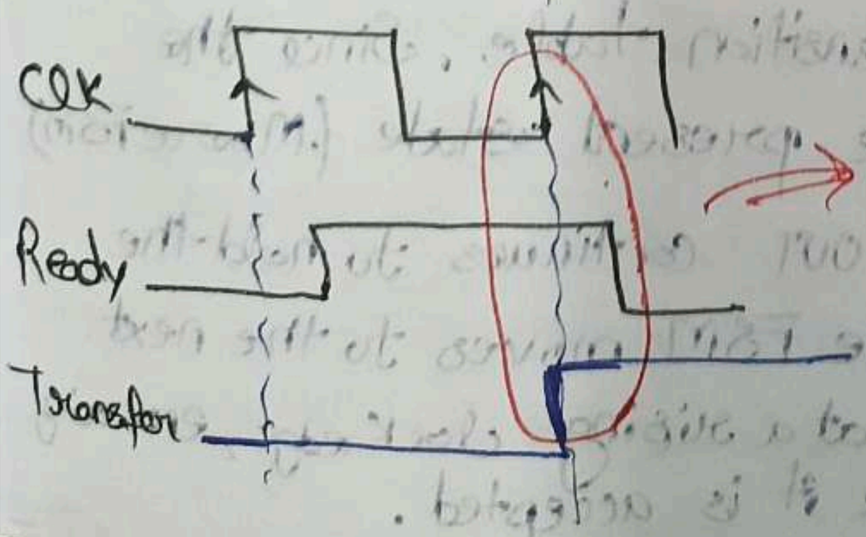
S_1 ⇒ A is to be transmitted to the receiver.

S_2 ⇒ B is to be transmitted to the receiver.

S_3 ⇒ C is to be transmitted to the receiver.

S_4 ⇒ DONE_STATE
Here, Valid = 0, Done = 1.

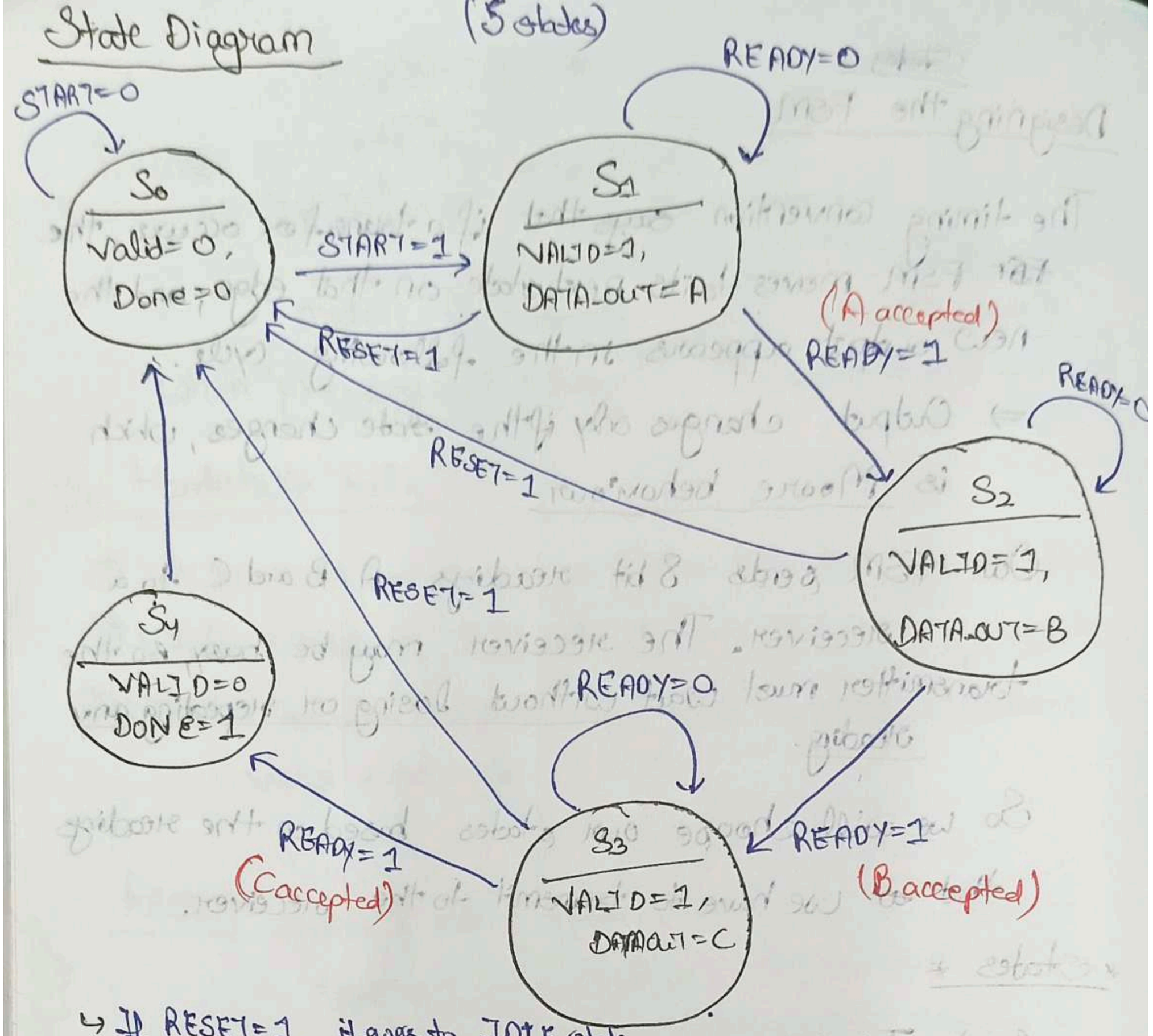
↳ All the transfer will happen only at the rising edge of the clock based on the value of READY.



→ Transfer will occur at the rising edge of clk. If READY = 1, i.e. receiver is ready to accept the data/reading.

State Diagram

(5 states)



↳ If RESET=1, it goes to IDLE state

↳ For states S_1, S_2, S_3, the input START is ignored, i.e. it is only used in IDLE state.

↳ For transition $S_4 \rightarrow S_0$, it is unconditional; i.e. it happens the next clock cycle occurs. (Handling Completion)

↳ DONE = 1 only when C is accepted, i.e. all three Data readings are accepted.

* Handle waiting

When READY = 0, our FSM remains in the same sending state (S_1, S_2 or S_3), as shown by the self loops in the state diagram and transition table. Since the outputs depend only on the present state (Moore FSM), VALID remains 1 and DATA_OUT continues to hold the current reading unchanged. The FSM moves to the next state only when READY = 1 at a rising clock edge, ensuring that no reading is changed before it is accepted.

State Transition Table

Present State	Input		Next State	Output		
	START	READY		VALID	DONE	DATA_OUT
IDLE(S_0)	0	x	IDLE (S_0)	0	0	x
S_0	1	x	S_1	1	0	x
S_1	x	0	S_1	1	0	A
S_1	x	1	S_2	1	0	A
S_2	x	0	S_2	1	0	B
S_2	x	1	S_3	1	0	B
S_3	x	0	S_3	1	0	C
S_3	x	1	S_4	1	0	C
S_4(DONE) -STATE	x	x	S_0	0	1	x

* Whenever the RESET is going HIGH (RESET=1), the FSM will go to IDLE state irrespective of the i/p. on present state. (Handling Reset)

Output Table

State	Valid	DATA_OUT	DONE
S_0	0	Don't care	0
S_1	1	A	0
S_2	1	B	0
S_3	1	C	0
S_4	0	Don't care	1

↳ DATA_OUT is A for S_1, B for S_2, C for S_3. It stays on that reading for as long as the FSM stays in that state.

↳ DONE is only high in the state S_4 (DONE-STATE). It always moves to IDLE in next clock edge.

∴ DONE is High exactly for one clock cycle.

VALID=0 in that same state.

* Timing Exercise

Rising Edge	Ready (READY)	Reading on DATAOUT	Transfer?
1	0	A	No
2	1	A	Yes
3	0	B	No
4	0	B	No
5	1	B	Yes
6	1	C	Yes

* Finding the BUG

VALID=0

If we set ~~valid=0~~ whenever READY=0 (i.e our VALID is dependent on READY)

In this case, The Transmitter has a reading ~~READY~~ ready to send, but READY=0, so it will set VALID=0.

↳ And receiver looks for VALID=1 before raising READY=1 (If it sees 0, it will always stays READY=0

* Therefore the transmitter will never accept the reading.

What controller should do??

↳ We have to keep the VALID dependent on state only. so that VALID=1 (state S_0, S_1, S_2), the receiver will see and can raise READY.

↳ While READY=0 (i.e. data is being accepted by the receiver in that case FSM stays in same state and DATA_OUT is unchanged.